Two-Stage Fiber and Axon Segmentation for TEM Micrographs

TEM Axon & Myelin Segmentation Challenge | Method Description

Abstract. We describe a two-stage pipeline for paired instance segmentation of axons and myelin in transmission electron microscopy (TEM). A fine-tuned Cellpose model first detects complete fibers at reduced resolution. Its instance map is restored to native resolution, and each fiber defines a crop of original TEM pixels. A compact U-Net receives that crop and the fiber mask as two input channels and predicts the inner axon. Myelin is reconstructed as the fiber minus its axon. The implementation separates instance detection from compartment prediction, masks annotated artifacts during fiber training, and preserves native pixels for the second stage. We specify the model, losses, default training protocol, and submission encoding; quantitative benchmark results are not claimed.

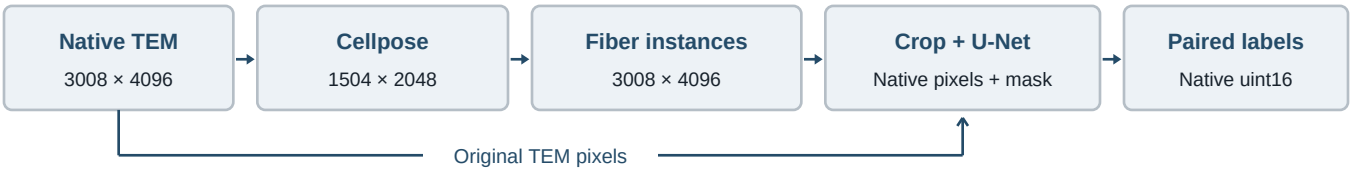


Fig. 1. Inference workflow. Shapes are (height, width). Fiber labels return to native resolution by nearest-neighbor interpolation; the U-Net obtains crop intensities directly from the original TEM image.

I. TASK AND REPRESENTATION

The objective is to identify individual myelinated fibers and assign each fiber an inner axon and an enclosing myelin compartment. We decompose this into whole-fiber instance detection followed by binary segmentation within each detected fiber. The outer prediction therefore supplies both an instance identity and spatial support for compartment prediction.

Let I be a native grayscale micrograph of size 3008 × 4096 and G its paired instance annotation. Following the challenge convention [1], zero denotes background, one denotes ignored artifacts, an even ID $a \geq 2$ denotes an axon, and $a + 1$ denotes its paired myelin. The complete ground-truth fiber is

$$F_a^{GT} = \{x : G(x) = a \text{ or } G(x) = a + 1\}.$$

Fiber targets are derived from original compartment labels during loading. A separate merged-mask dataset is not required. Artifacts remain identifiable through a validity map, even though their pixels are removed from the instance target.

A. Explicit image geometry

All shapes in this method use (height, width). Stage 1 receives 1504 × 2048 arrays, exactly half the native height and width. Images use area interpolation; integer labels and validity masks use nearest-neighbor interpolation. The code passes (width, height) only at the OpenCV resize boundary and never transposes an array to change its size.

Stage 2 operates on native-resolution crops. Restoring an instance map does not recover image detail lost by downsampling, so its crop pixels are read from I rather than from an enlarged Stage 1 input.

II. STAGE 1: FIBER INSTANCES

Stage 1 uses Cellpose 4.2.1.1 with its CPDINO initialization [2, 3, 5]. Cellpose predicts foreground logits and two spatial flow components, then groups pixels through its flow-based instance reconstruction [2]. Full resized micrographs are cached, while training uses random 512 × 512 crops. Normalization follows the pinned Cellpose defaults [3].

A. Artifact-masked supervision

Let $V(x) = 1$ when the original annotation is not artifact ID 1, and zero otherwise. V is resized with nearest-neighbor interpolation alongside the target. During augmentation, it is appended as an additional image channel so that image pixels, flows, and validity receive the same geometric transform. This channel is removed before network inference. Only transformed validity values greater than 0.999 are supervised, excluding mixed artifact edges and synthetic padding.

With $N = \Sigma_x V(x)$, target flow d, predicted flow $\hat{d}$, foreground target y, and foreground logit z, the implemented loss is

$$L_{fiber} = L_{flow} + L_{fg},$$

$$L_{flow} = (1 / 4N) \Sigma_x V(x) \|\hat{d}(x) - 5d(x)\|^2,$$

$$L_{fg} = (1 / N) \Sigma_x V(x) \, \mathrm{BCEWithLogits}(z(x), y(x)).$$

The factor 5 rescales target flows, and the flow weighting matches the pinned Cellpose training loss. Both flow regression and foreground classification exclude artifacts. A crop with no valid supervision is skipped instead of being used as a negative example.

III. STAGE 2: AXON SEGMENTATION

A. Native crops and mask conditioning

For each training fiber, we find the tight bounding box of the union of its GT axon and myelin. If that box has height h and width w, it is expanded on each side by ceil(0.15h) rows and ceil(0.15w) columns, then clipped to the micrograph boundary. Border fibers are retained. Fibers with a missing annotation partner are skipped and recorded in an annotation-issues table.

The first input channel contains original TEM intensities from this expanded box, including surrounding context. The second is the binary mask of this one fiber; neighboring fibers are not included in that channel. Integer images are divided by their dtype maximum (255 for uint8). Floating images must already lie in [0, 1].

Crops are never resized for the U-Net. They are padded on the bottom and right to multiples of 32: image values use edge replication and fiber masks use zeros. Predictions on padding are discarded. The same normalization, context rule, and padding function are shared by training and inference.

B. Compact U-Net architecture

The binary predictor follows an encoder-decoder U-Net [4]. Five encoder levels use the channel widths in Table I. Each block contains two bias-free 3 × 3 convolutions, each followed by GroupNorm with four groups and ReLU. Four max-pooling operations reduce spatial size. The decoder uses bilinear interpolation, concatenated encoder skip features, and matching convolution blocks. A 1 × 1 head produces one axon logit.

Table I. U-Net configuration at the default base width.

Component	Configuration
Input	TEM crop + target fiber (2 channels)
Encoder widths	16, 32, 64, 128, 256
Decoder widths	128, 64, 32, 16
Block	Conv-GN-ReLU, repeated twice
Output	1 logit; sigmoid at inference
Parameters	1,963,953 at base width 16

C. Augmentation and crop batching

Training samples receive random quarter-turn rotations and independent horizontal and vertical flips, each with probability 0.5. Spatial transforms are shared by pixels and labels. Pixel-only augmentation uses contrast factors 0.9-1.1, brightness offsets -0.05 to 0.05, clipping to [0, 1], and gamma 0.85-1.15. Crops are bucketed by longest side in 128-pixel intervals. Default batches contain eight fibers; buckets with any side above 700 pixels use batch size one.

IV. TRAINING PROTOCOL

A. Supervision and data partitioning

The two models are optimized independently. Stage 2 learns from GT fiber masks and does not require Cellpose predictions for training. For new runs, a single persistent micrograph split is shared by both stages. The default partition is approximately 70% training, 15% validation, and 15% held out, generated from sorted micrograph IDs with seed 42. Whole micrographs, rather than individual crops, are assigned to a split. Only train and validation images are loaded by the trainers.

The split CSV can instead be supplied explicitly and must cover the dataset exactly. Reusing existing experimental weights preserves their original training history; it does not make their historical splits equal to the new shared protocol.

B. Axon objective

Let F and A denote the GT fiber and axon in a crop, and let p = sigmoid(z). The U-Net loss is binary cross-entropy averaged over F plus a soft Dice loss. The following terms are computed per fiber and then averaged across the batch:

$$L_{axon} = L_{BCE,F} + 1 - D_{soft,F},$$

$$L_{BCE,F} = \Sigma_{x \in F} \text{BCEWithLogits}(z(x), A(x)) / |F|,$$

$$D_{soft,F} = [2\Sigma_{x \in F} p(x)A(x) + \varepsilon] / [\Sigma_{x \in F} p(x) + \Sigma_{x \in F} A(x) + \varepsilon],$$

where $\varepsilon = 10^{-6}$. Outside-fiber pixels and batch padding do not contribute to this objective.

C. Default optimization settings

Table II records configuration defaults, not measured outcomes or the verified training history of any supplied checkpoint. Each stage selects its best weights by the lowest validation loss.

Setting	Fiber / Axon
Optimizer	AdamW / AdamW
Learning rate	10^{-5} / 10^{-3}
Weight decay	0.1 / 10^{-4}
Epoch limit	100 / 50
Batch size	1 image crop / 8 fiber crops
Selection	Minimum validation loss

The fiber schedule uses linear warm-up over ten epochs and, for the 100-epoch default, halves the rate every five epochs during the final fifty. Scale jitter uses Cellpose's scale_range = 0.5. The U-Net plateau scheduler uses factor 0.5 and patience 3; early stopping uses patience 10. Gradients are clipped to norm 1; CUDA uses automatic mixed precision.

V. INFERENCE AND SUBMISSION OUTPUT

A. Fiber restoration and compartment prediction

Inference starts from an original 3008 × 4096 micrograph. Its area-downsampled 1504 × 2048 version is segmented by the fine-tuned Cellpose model, using tiled inference. The resulting integer instance map is restored to native shape using nearest-neighbor interpolation. IDs are never interpolated as continuous values.

For each predicted fiber F_i, the inference code constructs a native crop with the same 15% context rule used for GT training crops. The U-Net receives native image pixels and this predicted fiber mask. After removing padding, the output is thresholded at 0.5 and restricted to F_i:

$$A_i = F_i \cap \{x : p_i(x) \geq 0.5\},$$

$$M_i = F_i \setminus A_i.$$

Thus the two predicted compartments are disjoint and their union is the predicted fiber. This construction preserves the outer support and instance identity; it does not guarantee a connected axon, a nonempty myelin ring, or an anatomically correct outer boundary. Crops are processed individually, without resizing.

The default Cellpose inference settings are listed in Table III. No test-time augmentation, shape-fitting procedure, or extra learned postprocessing is added by the submission pipeline.

Table III. Default Cellpose inference settings.

Parameter	Value
Tile side / overlap	512 pixels / 0.25
Tile batch size	1
Flow iterations	1200
Flow threshold	0.4
Cell probability threshold	0.0
Minimum object area	2000 Stage 1 pixels
Diameter override	None

B. Paired instance export

For consecutive fiber index $k \geq 1$, the output assigns label $2k$ to its axon and $2k + 1$ to its myelin; background is zero. A native single-channel uint16 TIFF or PNG retains the input filename. The code checks the 32,767-fiber encoding limit before casting, preventing silent overflow. GT labels and artifact maps are not read during prediction.

VI. REPRODUCIBILITY AND SCOPE

The submission exposes train.py, predict.py, and download_weights.py, with paths and settings in config.py. Cellpose is pinned to 4.2.1.1; PyTorch and torchvision are pinned to 2.6.0 and 0.21.0. DINOv3 is pinned to a Git revision in requirements.txt. The U-Net definition and preprocessing utilities are shared between training and inference.

Training stores the split and configuration and publishes the checkpoint selected by validation loss. Cellpose resume sidecars bind optimizer and RNG state to the saved weights by checksum. The U-Net stores completed-epoch optimizer, scheduler, scaler, and RNG state. Multi-worker augmentation and GPU kernels need not be bit-identical across restarted processes or hardware.

Before prediction, both local checkpoint files are checked. Missing files are downloaded with gdown from user-configured Google Drive links. Optional SHA256 checks verify their identity, and the prediction manifest records checkpoint hashes, configuration, output filenames, and completion status.

A. Limitations and evaluation scope

A missed Stage 1 fiber cannot be recovered by Stage 2, and merged or split instances propagate to crop construction. Because the U-Net is trained on GT fiber masks but receives predicted masks at inference, robustness to outer boundary errors remains an empirical question. Axon clipping also prevents recovery of true axon pixels outside a predicted fiber.

A full assessment should pair fibers one-to-one by their complete-fiber IoU, report missed and unmatched instances separately, and measure axon/myelin Dice, HD95, and g-ratio error on matches [1]. For area-derived equivalent diameters, $g_i = \sqrt{(|A_i| / |F_i|)}$, with each prediction using its own fiber area. This document reports the implemented method and defaults; no benchmark accuracy, runtime, or hardware performance measurements are available here.

REFERENCES

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[4] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," *MICCAI*, pp. 234-241, 2015. arxiv.org/abs/1505.04597.

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